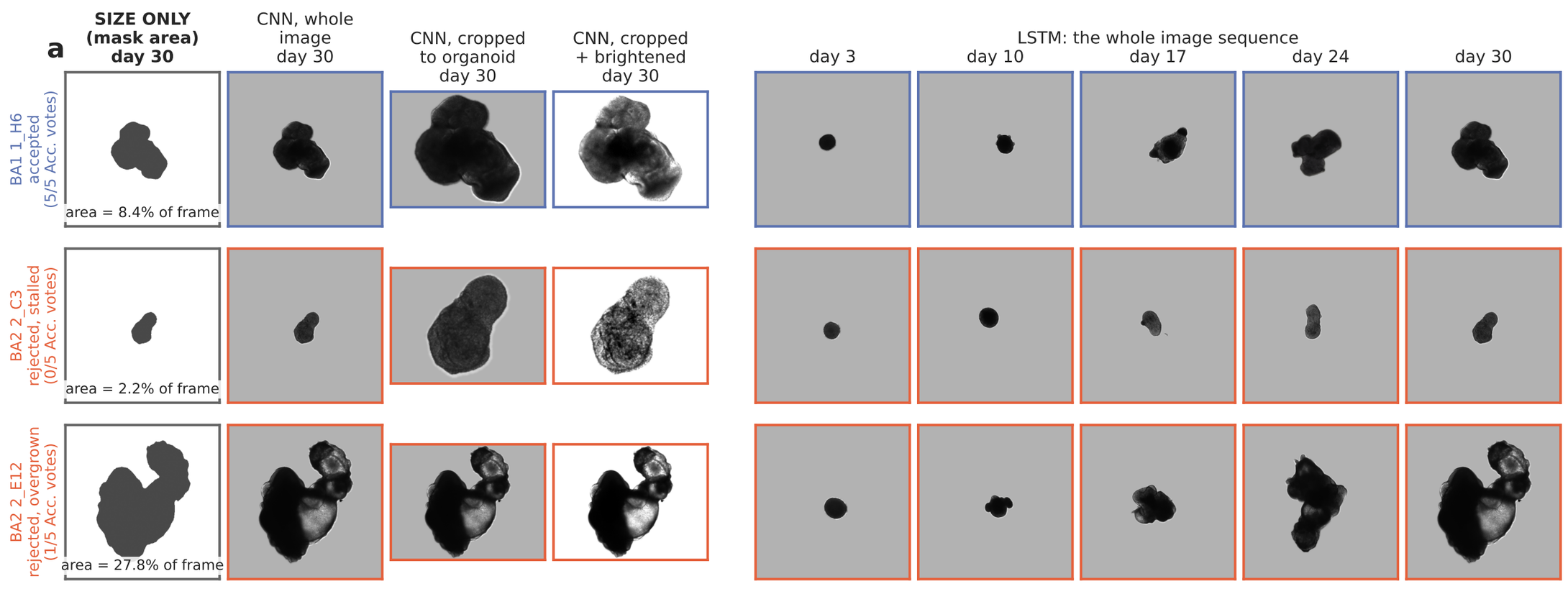
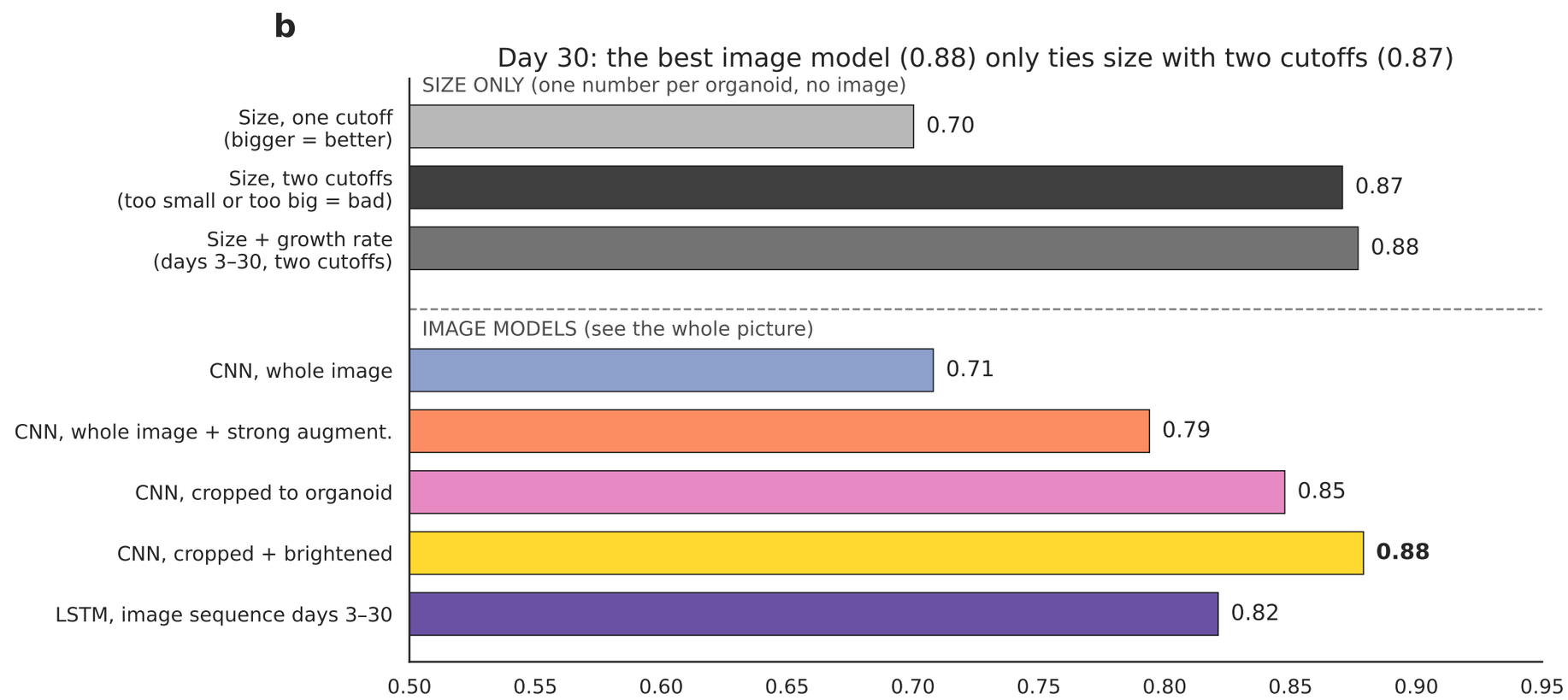
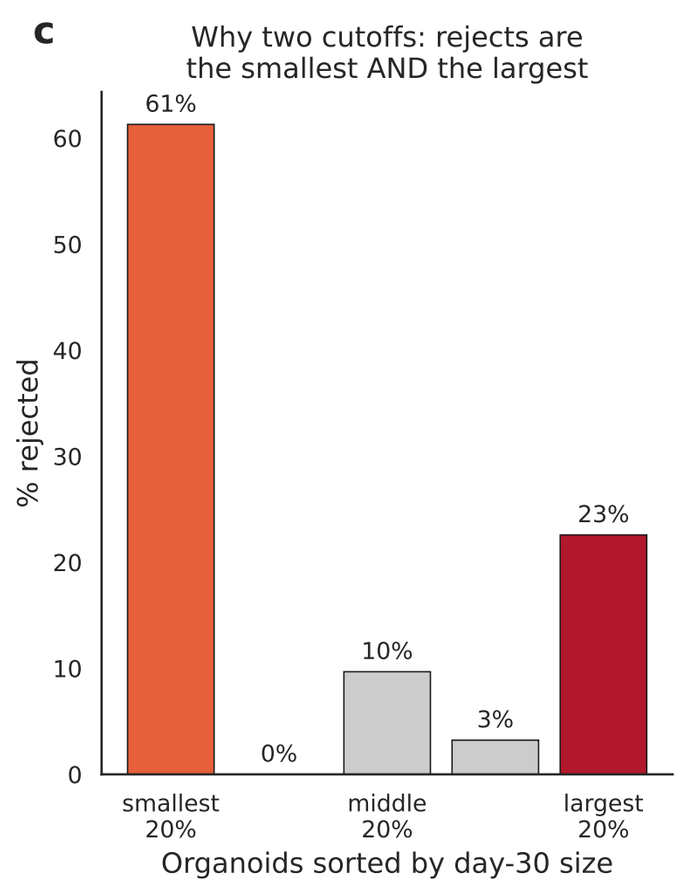
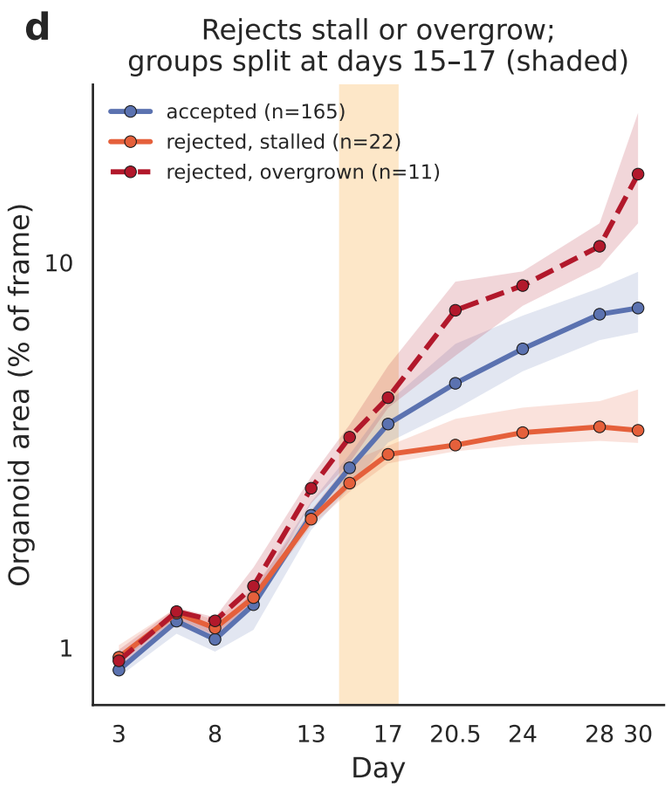
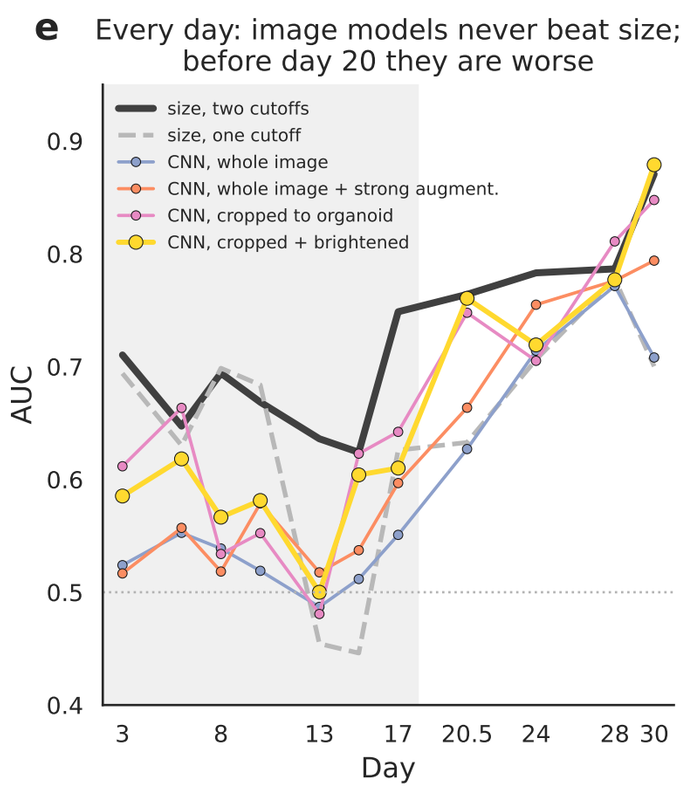
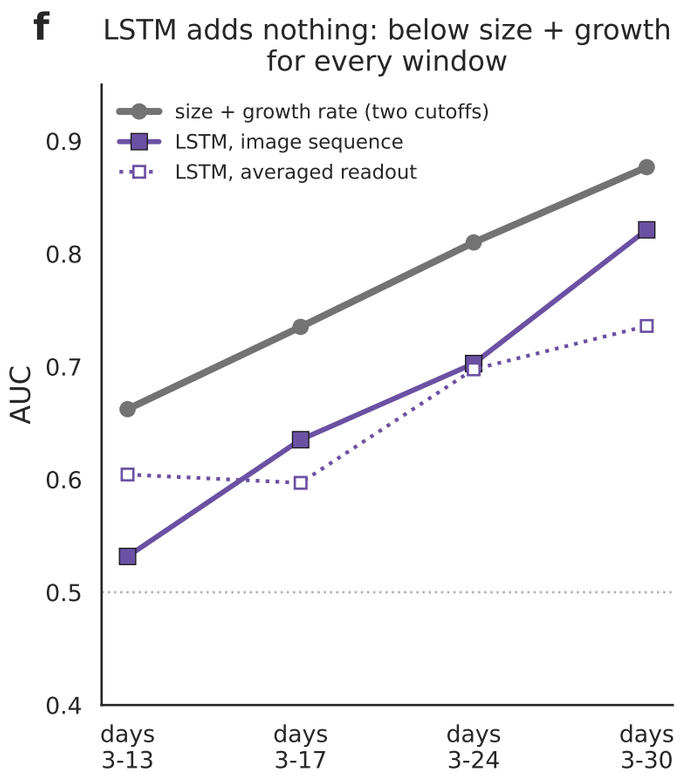

a
SIZE ONLY (mask area) day 30
CNN, whole image day 30
CNN, cropped to organoid day 30
CNN, cropped + brightened day 30
LSTM: the whole image sequence
day 3
day 10
day 17
day 24
day 30
BA1 1_H6 accepted (5/5 Acc. votes)
area = 8.4% of frame
BA2 2_C3 rejected, stalled (0/5 Acc. votes)
area = 2.2% of frame
BA2 2_E12 rejected, overgrown (1/5 Acc. votes)
area = 27.8% of frame


b
Day 30: the best image model (0.88) only ties size with two cutoffs (0.87)
SIZE ONLY (one number per organoid, no image)
Size, one cutoff (bigger = better)
0.70
Size, two cutoffs (too small or too big = bad)
0.87
Size + growth rate (days 3–30, two cutoffs)
0.88
IMAGE MODELS (see the whole picture)
CNN, whole image
0.71
CNN, whole image + strong augment.
0.79
CNN, cropped to organoid
0.85
CNN, cropped + brightened
0.88
LSTM, image sequence days 3–30
0.82
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
AUC at day 30 (0.5 = chance; mean of 4-fold × 10-repeat CV, same 155 organoids)


c
Why two cutoffs: rejects are the smallest AND the largest
% rejected
61%
0%
10%
3%
23%
smallest
middle
largest
20%
20%
20%
Organoids sorted by day-30 size


d
Rejects stall or overgrow; groups split at days 15–17 (shaded)
accepted (n=165)
rejected, stalled (n=22)
rejected, overgrown (n=11)
Organoid area (% of frame)
10
1
3
8
13
17
20.5
24
28
30
Day


e
Every day: image models never beat size; before day 20 they are worse
size, two cutoffs
size, one cutoff
CNN, whole image
CNN, whole image + strong augment.
CNN, cropped to organoid
CNN, cropped + brightened
AUC
0.9
0.8
0.7
0.6
0.5
0.4
3
8
13
17
20.5
24
28
30
Day


f
LSTM adds nothing: below size + growth for every window
size + growth rate (two cutoffs)
LSTM, image sequence
LSTM, averaged readout
AUC
0.9
0.8
0.7
0.6
0.5
0.4
days 3-13
days 3-17
days 3-24
days 3-30